Вариант 8

Решение уравнения имеет вид

$$\mathbf{U}_{\xi 0} = \frac{1}{2\varepsilon^2}\left[\frac{\partial^2 P_0}{\partial\xi\partial\tau} + \frac{\partial^2 P_0}{\partial\xi\partial\tau}\bar{\Psi}(\zeta) + \frac{\partial P_0}{\partial\xi}\bar{\Phi}(\zeta)\right], \tag{1}$$

где

$$\bar{\Psi}(\zeta) = F_2(\varepsilon\zeta)D_1 - F_1(\varepsilon\zeta) - 2F_4(\varepsilon\zeta)D_2, \quad \bar{\Phi}(\zeta) = 2F_3(\varepsilon\zeta) - F_2(\varepsilon\zeta)D_2 - 2F_4(\varepsilon\zeta)D_1,$$

$$\bar{\Psi_1}(\zeta) = \int\limits_{\zeta}^{1} \bar{\Psi} d\zeta, \quad \bar{\Phi_1}(\zeta) = \int\limits_{\zeta}^{1} \bar{\Phi}(\zeta) d\zeta, \quad \varepsilon(\omega) = \sqrt{\Re/2},$$